Baseline Superposition Principles

$$B_{o1}(t) = B'(t) + B''(t) \tag{1}$$
$$B_{o2}(t) = B''(t) - B'(t) \tag{2}$$
$$B_c(t) = B''(t) \tag{3}$$

Interval 1: $t_1 \leq t < t_2$

$$B'(t) = \left(\frac{B'_{max}}{DT_s}\right) t \tag{4}$$
$$B''(t) = \left(-\frac{B'_{max} n V_o}{V_{in_max} D T_s}\right) t + B'_{max}\left[1 - \frac{nV_o}{V_{in_max} D}(0.5 - D)\right] \tag{5}$$
$$B_{o1}(t) = \frac{B'_{max}}{DT_s}\left(1 - \frac{nV_o}{V_{in_max}}\right) t + B'_{max}\left[1 - \frac{nV_o}{V_{in_max} D}(0.5 - D)\right] \tag{6}$$
$$B_{o2}(t) = -\frac{B'_{max}}{DT_s}\left(1 + \frac{nV_o}{V_{in_max}}\right) t + B'_{max}\left[1 - \frac{nV_o}{V_{in_max} D}(0.5 - D)\right] \tag{7}$$
$$B_c(t) = B''(t) \tag{8}$$

Interval 2: $t_2 \leq t < t_3$

$$B'(t) = \left(\frac{B'_{max}}{DT_s}\right) t \tag{9}$$
$$B''(t) = 0 \tag{10}$$
$$B_{o1}(t) = B'(t) \tag{11}$$
$$B_{o2}(t) = -B'(t) \tag{12}$$
$$B_c(t) = 0 \tag{13}$$

Interval 3: $t_3 \leq t < t_4$

$$B'(t) = \left(-\frac{B'_{max} n V_o}{V_{in_max} D T_s}\right) t + B'_{max}\left(1 + \frac{nV_o}{V_{in_max}}\right) \tag{14}$$
$$B''(t) = 0 \tag{15}$$
$$B_{o1}(t) = B'(t) \tag{16}$$
$$B_{o2}(t) = -B'(t) \tag{17}$$
$$B_c(t) = 0 \tag{18}$$

Interval 4: $t_4 \leq t < t_5$

$$B'(t) = \left(-\frac{B'_{max} n V_o}{V_{in_max} D T_s}\right) t + B'_{max} \left(1 + \frac{n V_o}{V_{in_max}}\right) \tag{19}$$

$$B''(t) = \left(\frac{B'_{max}}{D T_s}\right) t - \frac{0.5 B'_{max}}{D} \tag{20}$$

$$B_{o1}(t) = \frac{B'_{max}}{D T_s} \left(1 - \frac{n V_o}{V_{in_max}}\right) t + B'_{max} \left(1 + \frac{n V_o}{V_{in_max}} - \frac{0.5}{D}\right) \tag{21}$$

$$B_{o2}(t) = \frac{B'_{max}}{D T_s} \left(1 + \frac{n V_o}{V_{in_max}}\right) t - B'_{max} \left(1 + \frac{n V_o}{V_{in_max}} + \frac{0.5}{D}\right) \tag{22}$$

$$B_c(t) = B''(t) \tag{23}$$

Interval 5: $t_5 \leq t < t_6$

$$B'(t) = 0 \tag{24}$$

$$B''(t) = \left(\frac{B'_{max}}{D T_s}\right) t - \frac{0.5 B'_{max}}{D} \tag{25}$$

$$B_{o1}(t) = B''(t) \tag{26}$$

$$B_{o2}(t) = B''(t) \tag{27}$$

$$B_c(t) = B''(t) \tag{28}$$

Interval 6: $t_6 \leq t < t_7$

$$B'(t) = 0 \tag{29}$$

$$B''(t) = \left(-\frac{B'_{max} n V_o}{V_{in_max} D T_s}\right) t + B'_{max} \left[1 + \frac{n V_o}{V_{in_max} D}(0.5 + D)\right] \tag{30}$$

$$B_{o1}(t) = B''(t) \tag{31}$$

$$B_{o2}(t) = B''(t) \tag{32}$$

$$B_c(t) = B''(t) \tag{33}$$